

RULE-BASED PYTHON CHATBOT

30-Minute Student Task • Completed Solution

NAME	REGISTER NO.	SECTION / DEPARTMENT
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Objective

Build a simple conversational chatbot that continuously accepts user messages, recognizes greetings and common topics, responds appropriately, and exits when the user types "bye" or "exit".

Chatbot Concept — "TripMate"

TripMate is a friendly travel-assistant chatbot. It can respond to greetings and answer questions about destinations, packing, budget travel, local food, and travel safety. It uses flexible keyword matching so phrases such as "hello bot" and "hey hello" are recognized.

Python Program

```
print("=====")
print("      TRIPMATE TRAVEL CHATBOT")
print("=====")

name = input("Hey traveller! Before we begin, what's your name? ").strip()
print(f"Nice to meet you, {name}!")
print("I can help with destinations, packing, budget, food, and more.")
print("Type 'bye' or 'exit' whenever you want to leave.\n")

while True:
    message = input(f"{name}: ").lower().strip()
    words = message.split() # whole-word matching, so "this" won't match "hi"

    if message == "bye" or message == "exit":
        print(f"TripMate: Goodbye, {name}! Safe travels and happy journeys!")
        break

    elif "hello" in words or "hi" in words or "hey" in words:
        print(f"TripMate: Hello {name}! Where would you like to go today?")

    elif "destination" in message or "place" in message or "visit" in message:
        print("TripMate: Popular picks: Goa for beaches, Manali for mountains,")
        print("Jaipur for heritage, and Kerala for backwaters.")

    elif "pack" in message or "luggage" in message:
        print("TripMate: Pack light! Carry ID proof, a power bank, basic medicines,")
        print("comfortable shoes, and clothes suited to the weather.")

    elif "budget" in message or "cheap" in message or "cost" in message:
        print("TripMate: Save money by booking tickets early, travelling off-season,")
        print("staying in hostels or homestays, and using local transport.")

    elif "food" in message or "eat" in message:
        print("TripMate: Try local street food and regional dishes,")
        print("but choose clean, busy stalls to stay safe.")

    elif "safe" in message or "safety" in message:
        print("TripMate: Share your itinerary with family, keep copies of documents,")
        print("and avoid carrying too much cash.")

    elif "thank" in message:
        print(f"TripMate: You're welcome, {name}!")

    elif "how are you" in message:
        print("TripMate: I'm doing great and ready to plan your trip!")

    else:
        print("TripMate: I'm not sure about that yet.")
```

```
print("Try asking about destinations, packing, budget, food, or safety.")
```

Requirement Checklist

ITEM	IMPLEMENTATION	STATUS
1	Asks the user's name before the main conversation.	Done
2	Uses while True: to keep the conversation running.	Done
3	Uses input() to receive messages.	Done
4	Cleans input using .lower().strip().	Done
5	Recognizes hello, hi, and hey.	Done
6	Answers multiple topics: destinations, packing, budget, food, safety, thanks.	Done
7	Uses if / elif / else for decision making.	Done
8	Uses break for bye or exit.	Done
9	Uses the user's name in responses.	Done
10	Uses else to provide a useful response for unknown messages.	Done
Challenge	Uses the in operator for flexible greeting/topic matching.	Done

Suggested Test Run

TEST	INPUT	EXPECTED BEHAVIOUR
Name	Dudigam	Welcomes the user by name
Greeting	HELLO BOT	Greets the user: 'Hello Dudigam!'
Topic	Suggest a place to visit	Lists popular destinations
Packing	What should I pack?	Gives packing tips
Budget	Give me low budget tips	Gives money-saving travel tips
Unknown	Tell me a random fact	Shows the fallback (else) message
Exit	bye	Says goodbye and ends the program

Target Flow

INPUT -> PROCESS -> RESPOND -> REPEAT -> EXIT

Clean • Flexible • Traveller-friendly • Easy to demonstrate